

Automatic Hydroponic Plant Protection from rain

1. Introduction:

This project aims to design an automatic hydroponic plant cover that will shield the hydroponic plants from rain to maintain the nutrients by preventing it from leaching and overwatering. This report documents the entire progress of the project up to this point, before the final Project Result & Evaluation stage, detailing the product design, construction, testing, and troubleshooting of the circuit.

2. Work Completed to Date:

• Detailed Description of Progress:

Since the beginning of the project, we have made significant technical progress in both the electronic and mechanical aspects of the system. The goal of the project was to design an automatic rain protection cover for a hydroponic system, controlled via rain detection and motor actuation. Below are the key tasks that have been completed:

Design Finalization:

We completed the circuit schematic design using simulation software and finalized the PCB layout using a design tool compatible with our fabrication method. Simultaneously, the mechanical design for the actuator frame and roof structure was developed.

Component Procurement:

All major components required for the electronic and mechanical systems were identified and purchased. This includes resistors, the LM741 op-amp, BC547 transistor, relays, N20 DC motors, limit switches, diodes, connectors, potentiometers, and mechanical frame materials (e.g., PVC pipes).

Circuit Simulation and Breadboard Testing:

The circuit was initially simulated using the Falstad simulator to verify the behavior of the rain sensor detection system and relay actuation. The same circuit was then prototyped on a breadboard, confirming that the comparator correctly responded to varying resistance simulating wet and dry conditions.

PCB Fabrication and Assembly:

1. LM741 comparator.

2. Voltage divider using rain sensor and a $10k\Omega$ resistor.
3. Adjustable threshold using a 10k potentiometer.
4. BC547 transistor and flyback diode.
5. Relay and four N20 Gearbox DC motors.
6. Limit switches integrated to control motor stop positions.

Changes from earlier version:

1. R1 and R3 increased to 10k (from 1k).
2. DC Gearbox Motors increased to 4 units (from 1).
3. Added LM741 comparator; its non-inverting input connected to voltage divider (rain sensor + trimpot). Previously, the voltage divider directly biased the BC547 base.
4. Status LED moved to op-amp output (previously at BC547 collector).

These changes were made to improve accuracy of rain detection, reducing unnecessary current usage, increase mechanical strength to roll the tarp roof, and ensure more reliable output indication through the op-amp.

- **Component Status:**

1. All required components have been acquired.
2. All components have been tested during breadboard and PCB-level testing.
3. No major component failures were encountered. However, relay triggering required adjustment of the comparator threshold (solved with potentiometer tuning).
4. Additional PVC pipes and reinforcement materials were purchased after initial mechanical testing revealed structural flexibility.

- **Testing and Results (Up to this point):**

- 1) Testing Performed:

- a. Simulations in Falstad and on breadboard.
- b. Measurements of voltage across op-amp outputs and relay.
- c. Relay response and transistor saturation behavior confirmed.
- d. Full system test with motor actuation and limit switch stopping.

- 2) Results:

- a. Comparator switches at expected thresholds (based on $\sim 6V$ reference).
- b. Relay successfully triggered at sensor resistance below $\sim 5k\Omega$.

- c. Motors operated correctly, stopping at limit switch positions.
- d. Current draw remains below 1A (safe for 12V power supply).
- e. The final PCB circuit behaves identically to the simulated version.

- **Challenges Encountered and Solutions:**

A. Relay Not Triggering Properly:

Caused by mismatch in comparator voltage. Solved by adjusting resistor values and installing a 10k potentiometer for fine tuning.

B. Frame Instability:

Early frame was too weak. Reinforced with screws

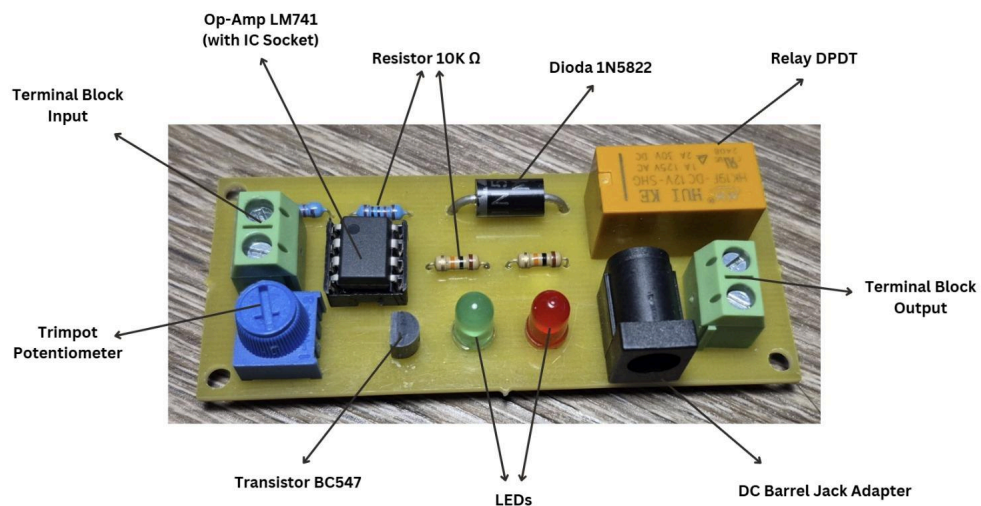
C. Limit Switch Misalignment:

Fixed by redesigning brackets in CAD and reprinting for better fit.

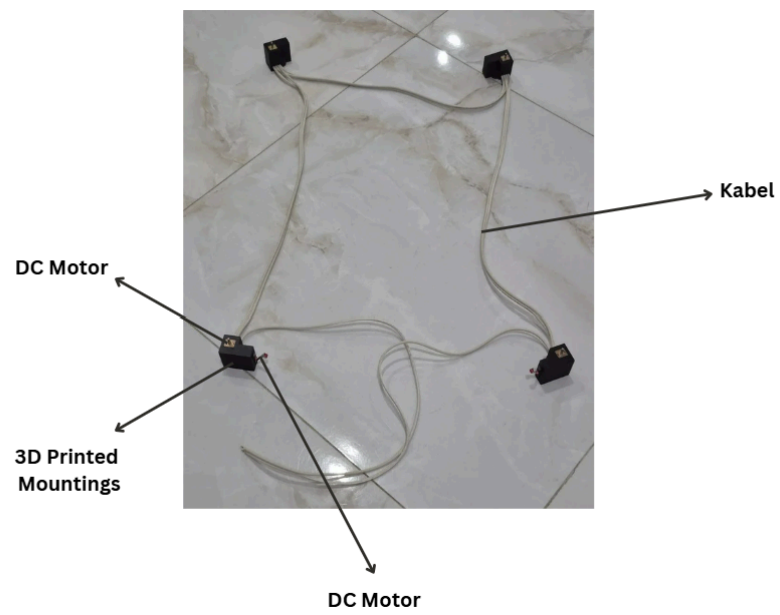
D. Motor Control Glitches:

Traced back to lack of flyback protection. Fixed by installing diodes across relay and motor terminals.

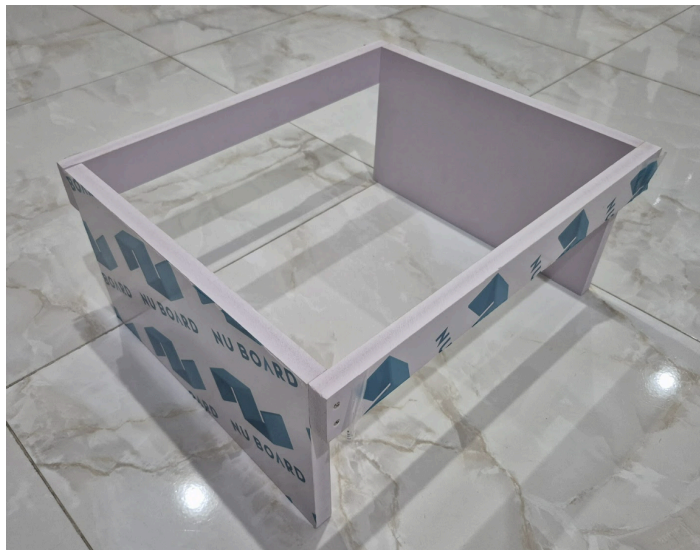
3. Visual Evidence:



Rangkaian Utama pada PCB



Rangkaian Limit Switch dan DC Motor



Bingkai Pot dan Atap Menggunakan PVC Foam Board



Memasang Rangkaian Limit Switch dan DC Motor pada Bingkai

4. Reflection and Planning:

- **Reflection on Progress (Entire Project to Date):**

Looking back on the project from the initial planning to the current integrated system, we can say that our progress has been steady and close to our expected timeline. Most milestones were achieved on time, though some stages—particularly mechanical assembly—took longer than anticipated due to adjustments in the frame design and component fit. We have gained significant experience in both circuit design and mechanical system integration. What went particularly well was our ability to validate our design through simulation (Falstad) and transfer it successfully to a functioning PCB. If there's one area we could have improved, it would be more detailed planning of the mechanical side in the early stages, as some of the redesigns could have been avoided.

- **Challenges and Lessons Learned (Entire Project to Date):**

Throughout the project, we faced several key challenges that became valuable learning moments. One challenge was fine-tuning the comparator circuit so that the relay would trigger consistently under varying rain sensor conditions. This taught us the importance of adjustability, which we solved by replacing fixed resistors with potentiometers. Another issue was the physical instability of the initial PVC frame, which required reinforcement and custom 3D-printed parts. We also encountered alignment problems with limit switches, which affected the actuator stopping mechanism. Through these challenges, we learned the importance of prototyping early and being flexible in our design approach. We also improved our teamwork and

problem-solving skills by dividing tasks efficiently and responding quickly to problems.

- **Current Status and Next Steps (Leading to Final Evaluation):**
Currently, the project is functionally complete. The circuit is fully operational and mounted on a PCB, the actuator system is assembled and integrated with the rain detection circuit, and the motors successfully respond to rain conditions by opening and closing the hydroponic cover. All sensors, switches, and relays have been tested.

What remains before the final evaluation:

- A. Conduct extended testing in real environmental conditions (e.g., water droplets, different rain intensities).
- B. Finalize and clean up wire management and mechanical mounting for presentation.
- C. Collect video/photo documentation and performance data (e.g., response time, current consumption).
- D. Prepare the final project report and presentation slides.
- E. Our goal for the final stage is to ensure the system is not only functional but also reliable and presentable, ready for demonstration during the final evaluation.

Progress documentation:

<https://drive.google.com/drive/folders/1GFRT7rV3YSRr9KsO9saTGnvedaAOEba6?usp=sharing>